

Theology Study Tracker — Application Specification

Version: 1.3 Date: May 2026 Status: Draft

Revision History

Version	Change
1.0	Initial draft
1.1	Renamed Module → Unit; added Notes system; added document references on work items; clarified Unit vs. Topic distinction
1.2	Confirmed Notes ↔ Work Items as explicit many-to-many; clarified Note parent vs. reference distinction
1.3	Added due dates / projected completion to Work Items, Units, Courses; calendar upgraded to planning view; Topics redesigned as owners and referencers; topic tagging added to all entities; Topics lose status/progress; cross-linking replaced by unified topic tag system; Notes tagged to Topics (many-to-many); Topic View defined

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1. Overview

The Theology Study Tracker is a personal, single-user web application for organizing, planning, and tracking self-directed theological study. It supports a hybrid organizational model combining structured course-based study (organized into units) with persistent topical study threads that accumulate knowledge over time. The application runs entirely on the user's local machine with no external dependencies or cloud services.

Goals

- Provide a unified place to manage courses, units, topics, readings, assignments, papers, and notes
 - Support both structured course-based study and freeform topical study through a consistent work item model
 - Plan and schedule study through due dates and projected completion dates, surfaced on a calendar
 - Track study habits including time spent, completion, and daily streaks
 - Capture and organize personal study methods and practices
 - Aggregate everything related to a subject in a single Topic View
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2. Deployment Architecture

Runtime Environment

The application is containerized using **Docker** and run locally via **Docker Compose**. No cloud hosting, authentication service, or external database is required.

Data Persistence

All application data is stored in a single **SQLite** database file exposed to the host machine via a **Docker bind mount** — a free feature of Docker Community Edition. The database file lives on the user's machine and persists independently of the container lifecycle.

Example `docker-compose.yml` structure:

```
yaml

services:
  app:
    build: .
    ports:
      - "3000:3000"
    volumes:
      - ./data:/app/data    # bind mount: ./data on host → /app/data in container
```

The SQLite file lives at `./data/theology.db` on the host. Copying that file is a complete backup of all application data.

Why SQLite

- Single file — easy to back up, inspect, and move
- No separate database container required
- Well-suited to single-user, local-first applications
- Fully supported by Docker bind mounts without any premium features

3. Scripture Reference Standard

All scripture references follow this format:

```
[3-char book abbreviation] [chapter]:[verse]
```

Examples:

Reference	Meaning
Gen 1:1	Genesis chapter 1, verse 1

Rom 8:28

Romans chapter 8, verse 28

Jhn 3:16

John chapter 3, verse 16

Rev 22:1-5

Revelation chapter 22, verses 1 through 5

Psa 119

Psalm 119 (whole chapter)

Ranges use an en dash (–) between verse numbers. A reference without a verse number implies the entire chapter. The application validates references against a known list of book abbreviations and flags malformed entries at input time.

4. Core Concepts

4.1 Unit vs. Topic

Unit (inside a Course)

A **Unit** is a bounded, sequential division within a course. It represents a defined body of work — a book, a subject area, or a phase of study — with a clear beginning and end in the course's scope. Units are the primary organizational structure for tracked, goal-oriented learning.

Use a Unit when:

- Working through a structured syllabus or self-designed curriculum
- The study is bounded by the course context
- Order and sequence matter (Unit 1 precedes Unit 2)

Example — Course: Hermeneutics

- Unit 1: *How to Read a Book* (Adler)
- Unit 2: *Exegetical Fallacies* (Carson)
- Unit 3: *Biblical Hermeneutics* (Zuck)

Topic

A **Topic** is a persistent study subject that accumulates content over time and across contexts. Topics are not bounded by any single course. They serve two roles:

1. **Owner** — a Topic can directly own work items and notes that are not part of any course.
2. **Tagger / Referencer** — any entity (Course, Unit, Work Item, Note, Study Session) can be tagged with one or more Topics, associating it with that subject regardless of where it lives

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Topics have no status and no progress tracking. They do not begin or end. The Topic View surfaces an informational count of complete vs. total work items associated with the topic, but this is not a tracked goal.

Use a Topic to:

- Organize freeform study not tied to any course
- Aggregate everything ever studied on a subject across courses, sessions, and notes
- Tag course work items so they appear in the relevant Topic View

Example — Topic: Hermeneutics This topic would surface:

- Work items it directly owns (standalone readings, exercises)
- Work items in the Hermeneutics course units tagged with this topic
- Units and courses tagged with this topic
- Study sessions tagged with this topic
- Notes tagged with this topic

4.2 Topic Ownership vs. Topic Tagging

These are two distinct relationships between topics and other entities:

Ownership applies only to Work Items and Notes. A topic *owns* a work item or note when that item has no Unit parent — the topic is its home. Owned items are created from within the Topic View and are listed there as primary content.

Tagging applies to all entities: Courses, Units, Work Items, Notes, and Study Sessions. A tag associates an entity with a topic for aggregation in the Topic View, without changing where the entity lives. A work item in a course unit can be tagged with multiple topics; it still belongs to the unit.

4.3 Work Item Parentage Rules

Every work item must satisfy at least one of the following:

- It belongs to a **Unit** (course context), or
- It is **owned by a Topic** (standalone context)

It may satisfy both. A work item in a unit may also be tagged with any number of topics. The application enforces this constraint at creation time: a work item cannot be saved without a unit parent or at least one owning topic.

Scenario	Unit Parent	Owning Topic	Additional Topic Tags
Course work item, no topic	✓	—	Optional
Course work item with topics	✓	—	✓ One or more
Standalone topic work item	—	✓ Required	Optional additional
Cross-context item	✓	✓	Optional additional

5. Data Model

Entities

Course

- Title
- Description
- Status (Active / Paused / Complete)
- Start date (optional)
- Target completion date (optional)
- Actual completion date (set automatically when marked Complete)
- Created date

Unit *(belongs to a Course)*

- Title
- Order within course
- Description
- Target completion date (optional)
- Actual completion date (set automatically when all work items complete)

Topic

- Title
- Description
- Type (Book of Bible / Theological Theme / Doctrine / Other)
- Parent topic (optional, one level deep)
- Created date

- *(No status, no target date, no progress tracking)*

Work Item

- Type (Reading / Assignment / Paper / Practice Session)
- Title
- Status (Not Started / In Progress / Complete)
- Estimated duration (minutes)
- Due date (optional)
- Completion date (set automatically when marked Complete)
- General notes (freeform plain text)
- Unit parent (nullable — null if owned by a topic)
- Owning topic (nullable — required if no unit parent; see Section 4.3)

Note

- Title
- Body (markdown-formatted text)
- Primary parent (exactly one: Course, Unit, Topic, Method, or Study Session)
- Created date
- Last modified date

Method

- Name
- Description / personal notes
- Created date

Study Session

- Date
- Duration (minutes)
- Associated work item (optional)
- Method used (optional)
- Reflection note (freeform plain text)
- Scripture tags (optional)

Scripture Tag

- Reference (formatted per Section 3)
- Associated entity (Work Item, Topic, or Study Session)

- Associated entity (work item, topic, or study session)

Relationships

Course —< Units —< Work Items (unit-owned)

Topic —< Work Items (topic-owned, no unit parent)

Work Items >—< Topics [many-to-many: tagging]

Notes >—< Topics [many-to-many: tagging]

Courses >—< Topics [many-to-many: tagging]

Units >—< Topics [many-to-many: tagging]

Study Sessions >—< Topics [many-to-many: tagging]

Notes >—< Work Items [many-to-many: work item references]

Note —o Primary Parent [exactly one of: Course, Unit, Topic, Method, Session]

Study Sessions —o Work Items [optional association]

Study Sessions —o Methods [optional association]

Scripture Tags —o Work Items / Topics / Sessions

Note on Topic Relationships

Topics participate in three kinds of relationships:

1. **Ownership of work items** — a work item with no unit parent is owned by a topic (FK on work item table).
2. **Ownership of notes** — a note whose primary parent is a topic is owned by that topic.
3. **Tagging** — any entity can be tagged with any number of topics via a join table (one per entity type: `course_topics`, `unit_topics`, `work_item_topics`, `note_topics`, `session_topics`).

6. Functional Requirements

6.1 Study Structure — Courses and Units

- A course contains one or more **units**, ordered sequentially.
- Each unit contains one or more **work items**.
- Courses have a status: Active, Paused, or Complete.
- Courses have optional start and target completion dates. Marking a course Complete sets the actual completion date automatically.

the actual completion date automatically.

- Units have an optional target completion date. A unit is automatically marked complete when all its work items are complete; its actual completion date is set at that moment.
- A course displays an overall progress percentage (completed work items / total work items across all units).
- Units can be reordered via drag-and-drop or a manual order field.
- Courses and Units can each be tagged with any number of Topics.

6.2 Study Structure — Topics

- A topic has a title, description, type (Book of Bible / Theological Theme / Doctrine / Other), and an optional parent topic (one level deep for subtopics).
- Topics have no status, no target date, and no progress goal. They do not begin or complete.
- Topics display an informational count of complete vs. total work items associated with them (both owned and tagged), for reference only.
- Work items and notes can be created directly within a Topic View, in which case the topic becomes their owner.
- Topics can be tagged onto Courses, Units, Work Items, Notes, and Study Sessions regardless of where those entities live.

6.3 Notes

Notes are markdown-formatted documents serving as the primary medium for capturing study content, written work, commentary, and reflections.

Note Properties

Field	Description
Title	Short descriptive name
Body	Markdown-formatted text, rendered in the UI
Primary parent	Exactly one: Course, Unit, Topic, Method, or Session
Topic tags	Zero or more Topics (many-to-many, independent of primary parent)
Created / Modified	Timestamps, managed automatically

Parent Attachment

- Every note has exactly one primary parent, which determines where it is listed and managed in the UI.

- A note can be moved to a different primary parent via an explicit reassignment action.

Topic Tagging on Notes

- Any note can be tagged with any number of Topics, regardless of its primary parent.
- A note parented to a Unit (e.g., lecture notes on Calvin's view of free will) can be tagged with Topics "Free Will," "Salvation," and "Problem of Pain," causing it to surface in all three Topic Views.
- Topic tags on notes are managed from within the note editor via a tag input field.

Work Item References (Many-to-Many)

- A note can be referenced by any number of work items of type Assignment, Paper, or Practice Session.
- A work item can reference any number of notes.
- References are managed from the work item view (attach/detach notes by search or browse).
- Deleting a note referenced by work items requires a confirmation warning. References are removed on deletion; work items are not affected.

Editor Requirements

- The note editor supports live markdown preview (split-pane or toggle).
- Supported markdown: headings, bold, italic, blockquote, ordered and unordered lists, inline code, code blocks, horizontal rules, links.
- Notes auto-save while editing. A last-saved timestamp is displayed.
- The topic tag input is accessible within the editor view.

Note Listing

- Each Course, Unit, Topic, Method, and Session view includes a Notes section listing attached notes with title, last modified date, topic tags, and a one-line content preview.
- Notes can be created, edited, and deleted from within their primary parent's view.

6.4 Work Item Types

All work items share these common fields:

Field	Description
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Title	Short descriptive name
Status	Not Started / In Progress / Complete
Estimated duration	In minutes
Actual time logged	Derived from associated study sessions
Due date	Optional target date for completion
Completion date	Set automatically when marked Complete
General notes	Freeform plain text for brief annotations
Scripture tags	Zero or more references
Topic tags	Zero or more Topics (many-to-many)
Unit parent	The unit this belongs to (null if topic-owned)
Owning topic	The topic that owns this item (null if unit-owned; required if no unit parent)

Reading

Field	Description
Source	Book title, article name, or "Scripture"
Author	Optional
Location	Page range (for books/articles) or scripture reference
Format	Physical Book / PDF / Online Article / Scripture

Readings do not use the note reference system.

Assignment

Field	Description
Description	What the assignment requires
Referenced notes	Zero or more Note entities representing the deliverable

Paper

Field	Description
Prompt or topic	The question or subject being addressed
Word count target	Optional
Score or grade	Optional, freeform (letter grade, percentage, or narrative)
Referenced notes	Zero or more Note entities (outline, draft, final, sections)

Practice Session

Field	Description
Method used	Reference to a Method (see Section 6.5)
Scripture passage	The text studied, in standard format
Duration	In minutes
Referenced notes	Zero or more Note entities representing the written output

6.5 Methods & Practices

- The user can create named study methods (e.g., Inductive Bible Study, SOAP Method, Lectio Divina, Grammatical-Historical Analysis).
- Each method has a title, description, created date, and a Notes section for detailed personal notes on applying the method.
- Any Practice Session work item or Study Session can reference a method.
- The application tracks how often each method has been used, displayed as a usage count with a linked session list.
- Methods are not deleted when referenced sessions are deleted; the reference is nullified instead.

6.6 Progress Tracking

Completion

- Every work item has a completion checkbox.
- Marking complete sets the completion date automatically. Un-completing clears it.
- A unit is automatically marked complete when all its work items are complete.
- A course may be manually marked complete; doing so records the actual completion date.

Time Logging

- Time is logged via Study Sessions.
- Sessions can be attached to a specific work item or logged free-form.
- Sessions require date and duration in minutes. Method, reflection, and topic tags are optional.
- Total logged time is displayed at the work item, unit, course, and topic level.

Study Streaks

- A streak is the number of consecutive calendar days on which at least one study session was logged.
 - The current streak and all-time longest streak are both displayed.
 - A streak breaks if no session is logged for a full calendar day, based on local system date.
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6.7 Calendar & Planning View

The Calendar is a combined retrospective and planning view. It shows both what has been done and what is scheduled, giving the user a complete picture of past activity and upcoming commitments.

Past (Retrospective Layer)

- Study session dates are shown as logged activity, shaded by total minutes (four-level intensity, consistent with the dashboard heatmap).
- Work item completion dates are marked on their respective days.
- Clicking a past day shows all sessions logged and items completed on that date.

Future (Planning Layer)

- Due dates for Work Items appear as scheduled markers on their target date.
- Target completion dates for Units appear as milestone markers.
- Target completion dates for Courses appear as milestone markers.
- Overdue items (due date in the past, not yet complete) are visually distinguished from upcoming items.

View Modes

- **Month view** — default. Shows the full calendar month with session intensity, completion markers, and due date markers.
- **Week view** — shows a single week in greater detail, listing each work item due and each session logged per day.
- **Agenda view** — a chronological list of upcoming due dates and milestones, starting from today, with no fixed time grid.

Interaction

- Clicking a future due date marker navigates to the associated work item, unit, or course.
 - Clicking a past completion marker navigates to the associated work item.
 - Clicking a past session marker opens the session log for that day.
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6.8 Dashboard

The dashboard is the application's home view.

Required elements:

- Current streak and all-time longest streak
 - Last session date and duration
 - Total time studied this week and this month
 - Active courses (up to 5) with title, completion percentage, and target completion date if set
 - Active topics (up to 5) with title and associated item count
 - Upcoming due dates — next 7 days across all work items, units, and courses
 - Recently modified work items (last 5) with direct links
 - Recently modified notes (last 5) with direct links
 - Calendar heatmap (condensed, trailing 3 months, with future due date indicators)
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6.9 Topic View

Each topic has a dedicated view that aggregates all content associated with it, regardless of where that content lives in the rest of the application.

The Topic View displays:

Section	Contents
Owned Work Items	Work items for which this topic is the primary owner (no unit parent), grouped by type
Tagged Work Items	Work items in courses/units that are tagged with this topic
Tagged Courses & Units	Courses and units tagged with this topic
Tagged Notes	Notes tagged with this topic, from any primary parent
Tagged Sessions	Study sessions tagged with this topic
Subtopics	Child topics (one level deep), each linking to their own Topic View

Informational summary (not a tracked goal):

- Total work items associated (owned + tagged): X complete of Y total
- Total time logged on associated sessions

Actions available from the Topic View:

- Create a new work item owned by this topic
- Create a new note owned by this topic
- Tag existing entities with this topic (via search/browse)
- Remove topic tags from entities
- Link to / unlink from a parent topic

6.10 Scripture Reference Tagging

- Tags are entered in the standard format defined in Section 3.
- The application validates format at entry and rejects malformed references.
- A **Reference View** allows browsing by scripture reference, returning all entities tagged with that reference or within a chapter range.
- Range lookup: entering Rom 8 returns all items tagged with any verse in Romans 8.

6.11 Search

A global search bar is accessible from all views.

- Searches across: course titles, unit titles, topic titles and descriptions, work item titles, work item notes, note titles, note body text, session reflections, and scripture tags.
- Results are grouped by type: Courses/Units, Topics, Work Items, Notes, Sessions.
- Scripture reference search supports partial matching: `Rom 8` returns all items tagged with any verse in Romans 8.
- Search is entirely local; no network access required.

6.12 Data Export

Export is triggered manually from a Settings or Export page.

Format	Contents	Re-importable
JSON	Full structured export of all data and relationships, including all topic tags and many-to-many joins	Yes
Markdown	All notes organized by primary parent entity, plus session reflections	No

- JSON export must be fully round-trip re-importable, restoring all data and relationships including topic tags, note references, and session associations.
- No data is sent to any external service during export.

7. Non-Functional Requirements

Requirement	Specification
Single user	No authentication, user accounts, or multi-tenancy
Local only	No network requests except to serve the application within Docker
Offline capable	Fully functional without an internet connection
Data ownership	All data lives on the user's machine in a bind-mounted SQLite file
Backup	Copying <code>./data/theology.db</code> constitutes a complete backup
Performance	All views load in under 500ms on a standard local machine

Platform	Runs on any machine with Docker Desktop (Community Edition) installed

8. Out of Scope

Feature	Reason for Exclusion
Cloud sync / user accounts	Single-user, local-first design; adds complexity with no benefit
AI-generated commentary or study content	Pulls focus away from primary sources and personal methods; theological reliability is a concern
Bible reading plan generator	Separate concern; the course/unit structure already handles what to read
Sermon or teaching prep tools	Different workflow and purpose from personal study tracking
Gamification (badges, XP, leaderboards)	Trivializes the subject matter; streaks are sufficient habit reinforcement
Social sharing	Out of scope for a personal tool
Church management features	Entirely different application domain

End of Specification — v1.3